Manuel M. Muñoz Cárdenas

Computer Science Engineering Student — Yachay Tech University • Ecuadorian and Spanish citizen

manuel.munoz@yachaytech.edu.ec • manuel976431@gmail.com • Urcuquí, Ecuador

github.com/xMane13 • linkedin.com/in/manuelmoon • ORCID 0009-0002-0435-3631 • Google Scholar

Profile

Computer Science Engineering student working on applied machine learning and its rigorous evaluation, in computer vision and federated learning. Thesis on hierarchical federated learning over wireless networks on a packet-level platform of my own; two publications and four manuscripts in preparation. Interested in trustworthy, data-efficient vision-language models and distributed learning.

Education

B.Sc. in Computer Science Engineering 2020 – Mar. 2027 (expected)

Yachay Tech University, Urcuquí — GPA 8.50/10, 10th semester

- Relevant coursework: Computer Vision (9.4/10), Artificial Intelligence, Intelligent Agents, Natural Language Processing, Neural Networks, Cryptography, Machine Learning in Production, HPC.
- Studied Physics up to the 5th semester: numerical modeling, finite differences and finite elements.

High School Diploma — International Baccalaureate (IB) Diploma 2016 – 2020

Colegio de Bachillerato Machala

- IB Diploma 28/45 (Nov. 2019), highest score in the school's history; final high school grade 8.74/10.

Research

Undergraduate Thesis Researcher — Yachay Tech University 2026 – present

Performance Evaluation of Hierarchical Federated Learning Architectures in Networked Environments • Advisors: J. P. Astudillo León, L. Lemus Cárdenas

- Built a packet-level platform for flat and hierarchical FL in `ns-3`, with the neural network implemented natively in C++ inside the simulator, so every model update crosses simulated sockets, the 802.11 MAC and IP fragmentation; configurable trees, channel plans and transports, with every change verified byte for byte.
- Pre-registered, paired experiments on a real tabular dataset (39k records) covering synchronization, data heterogeneity, aggregation, transport, payload and scale: the synchronous all-or-nothing round and access to the medium, not the hierarchy itself, are what limit learning over lossy or crowded wireless links.
- Proposed Adaptive Upload Staggering: each edge sets its cell's upload window from its own MAC failures, without extra messages or knowledge of the scale. On a shared channel it raised completed rounds from 31% to 98% (4×4) and from 14% to 92% (8×4), and still recovered rounds under vehicular mobility.

Independent Research — Methodological Audit of Pap Smear Classification 2026

SIPaKMeD dataset (4,049 cells, 966 slides), PyTorch • github.com/xMane13/cervical-cancer-sipakmed

- Showed that random splitting leaks slide-level information and hides about 1 in 4 errors relative to slide-grouped cross-validation (McNemar $p = 0.012$).
- Linear probing and area-normalized Grad-CAM over all cells: the network attends to the nucleus but encodes its color, not its shape, which explains why 26 expert morphological descriptors hurt performance and why stain shifts cut accuracy by 20 points.

Research Assistant — FLEAD Project, Yachay Tech University Oct. 2026 – present

Federated Learning for Embedded and Distributed Systems • Supervisor: J. P. Astudillo León. FL on edge devices (Raspberry Pi class) and network and mobility simulators, for IoT, transportation and medical applications

Research Assistant — Yachay Tech University 2025

Socioeconomic inequality and the perception of democracy in Latin America, 1995–2023: reproducible ETL and ML models for classification and clustering • Advisor: E. Cuenca

FedParam-RAG — Federated RAG 2026

NLP final project with J. Tipán and F. Valenzuela • github.com/xMane13/FedParamRag

- **LoRA** adapters trained per silo, exchanging only parameters, never local text; privacy contract verified with `pytest`.

Math2Braille — Co-Founder and Developer 2023 – 2025
Translator from Spanish math PDFs to Nemeth Braille; basis of the CSEI 2023 paper and a 2023 poster award

Publications

1. Hurtado, E., Arteaga, L., Duta, J., **Muñoz, M.** *et al.* (2024). **Development of a Python Program for Converting Spanish Math PDFs Into Braille.** *Proceedings of CSEI 2023*, Lecture Notes in Networks and Systems, vol. 797, Springer, Cham, pp. 252–263. doi:10.1007/978-3-031-70981-4_18.
2. Ariza García, E., Arteaga, L., Duta, J., Hurtado, E., Moya, R. & **Muñoz, M.** (2024). **La Fórmula de Leibniz y el Truco Favorito de Feynman** [Leibniz's Formula and Feynman's Favorite Trick]. *Mixba'al, Revista Metropolitana de Matemáticas*, 15(1), pp. 73–94. doi:10.24275/uami/dcbi/mix/v15n1/eljerm.

Manuscripts in Preparation

Co-author, with E. Hurtado-Almeida *et al.* (W-HPC research line, Yachay Tech)

1. *One-Class Multimodal Geometric Intrusion Detection with Complexity-Penalized Structure Selection.*
2. *Online and Drift-Aware OC-MM-WHPC: Guarded Streaming Adaptation with Interpretable Alarm Response.*
3. *Open-World and Uncertainty-Aware OC-MM-WHPC: Rejection, Querying, and Guarded Feedback Under Streaming Context.*
4. *Native Explanation and Empirical Robustness Diagnostics for OC-MM-WHPC Intrusion Detection.*

AI and Systems Projects

Deepfake Detection on FaceForensics++ 2026
With A. Chávez • github.com/xMane13/Detection-DeepFake-FF-. Frame and face extraction (MTCNN); Xception binary and multiclass classifiers over four manipulation methods, with per-method ensembles

Intracranial Hemorrhage Detection on Head CT 2026
PyTorch, RSNA dataset: multichannel contrast inputs (CLAHE, gamma) vs. grayscale control, teacher-model pseudo-labeling

Quest Align / SpineTrack — 2nd Place at CompuFest 2026 2025 – 2026
Spinal posture analysis with Meta Quest 3S and IMU sensors: acquisition, synchronization and processing; with C. Castro

Teaching and Leadership

President — IEEE Computational Intelligence Society (CIS), Yachay Tech 2026 – present
Workshops, talks and technical events on AI and ML with IEEE EPS; organizer of Conecta YT, a national-scale job fair

Instructor — Preparatory C Programming Course (12 sessions) 2026
IEEE CIS / EPS: syllabus, materials and assessments; also academic mentor and private tutor since 2022

Full-Stack Developer — Kichwa Outreach Project 2025
Community engagement project supervised by R. Fonseca-Delgado: Kichwa learning platform (Django, React)

Co-Founder and Vice President — AEMCiC 2024 – 2025
Student association of the School of Mathematical and Computational Sciences: bylaws, planning and representation

Technical Skills

Languages: Python, C, C++, TypeScript/JavaScript, SQL • **Spoken:** Spanish (native), English (advanced)

Machine learning: PyTorch, scikit-learn, computer vision (OpenCV, MediaPipe, Grad-CAM), federated learning, LoRA, reinforcement and imitation learning, agents (LangGraph), RAG

Systems and web: Linux, Docker, Kubernetes, MPI/HPC, CUDA, Git, LaTeX, Next.js, React, Django

Networking and simulation: `ns-3`, SUMO, 802.11, mobility modeling, vehicular scenarios

References

Manuel Eugenio Morocho-Cayamcela, Ph.D.
Computer Vision; IEEE CIS • mmorocho@yachaytech.edu.ec

Diego H. Peluffo-Ordóñez, Ph.D.
Natural Language Processing • dpeluffo@yachaytech.edu.ec

Juan Pablo Astudillo León, Ph.D.
Thesis Advisor • jastudillo@yachaytech.edu.ec

Rigoberto Fonseca-Delgado, Ph.D.
Kichwa Outreach Supervisor • rfonseca@yachaytech.edu.ec